SOIL

19 January 2019

THE FIRST DROP OF RAIN

The moment the first drops of rain strike a dry soil surface and wetting begins, the activity of microorganisms in the soil peaks. They wake when they see water — and that wonderful smell is the scent of the substances they secrete. Geosmin. Petrichor. What we call the scent of earth is in fact the cry of joy of billions of living creatures.

> "Soil treats you the way you treat it, friends. Treat it badly, even without meaning to, and it gives you no harvest; it treats you badly in return."

Soil is not an inert substrate — it is a living system, an organism with memory. It harbours billions of microorganisms in every gram, and these living things are far older, far more resilient than human beings. Commercial nitrogen fertilisers produce nitrogen oxides — a greenhouse gas 300 times more powerful than carbon dioxide. Repeated chemical fertilising builds up salt compounds, lowering the soil's capacity to hold water. Excessive irrigation dissolves soil nutrients and causes salinisation — exactly what is happening in the GAP region. Soil returns what is done to it: cared for, it becomes fertile; neglected, it becomes barren. This is not metaphor but biochemical fact. The human-soil relationship is the microcosm of all ecological relations.

A voice that has chaired a branch of a professional chamber for fifteen years recounts a journey from ordinary citizen to trade union representative, to union management, to the chair of the chamber. Seeing how science is manipulated — noticing how certain truths are pushed aside while the slightest things are given a glossy polish — is what drew this voice into the anti-GMO struggle. The anti-GMO platform's struggle has been going on since 2004 — a militant resistance to corporate control of seed. The platform against water commodification was founded ahead of the 2009 World Water Forum, but because it brought together organisations with different funding sources and motivations, fragility was inevitable. Farmer organisation is a precondition for agricultural emancipation in Turkey — but cooperative organising has to confront structural inequality, not turn into an institutional front.

The figures are painful: in the last fifteen to sixteen years, Turkish farmers have abandoned 3.2 million hectares of agricultural land — an area larger than Belgium. This is not only loss of land; every abandoned field carries away knowledge accumulated over generations. Yet Turkey has 4 million hectares of arable land left fallow — an area the size of the Netherlands, holding food security potential but unused. Agricultural prices are set on global commodity markets: Chinese cotton undercuts Turkish cotton, the farm becomes unprofitable, the young population rejects the countryside — both from economic necessity and cultural drift. The chemical agriculture "revolution" of the 1960s promised yield increases — farmers became dependent on inputs they could not afford, and soil degradation deepened the dependency.

Activists must choose their battles — energy, water, seeds. Dispersing in all directions leads to platform fatigue and collapse — when political conditions change, that collapse accelerates. Long-term organising requires emotional resilience and material security; without these even the best intentions dissolve.

THERE IS NO ECOLOGY WITHOUT LABOUR

Research in ecological agriculture examines production but ignores labour conditions. If organic, sustainable production has meaning in a region while worker exploitation is rising, that sustainability is hollow — even deceptive.

Pesticide exposure directly affects workers' health, but the heaviest burden of that exposure is borne by workers in the lowest income bracket, the least protected. Labour justice is an ecological matter — and ecology is incomplete without labour justice.

> "Why are seasonal agricultural workers from Cizre, Şırnak? Why there? Most of the workers in Cizre settled in the town centre after being forcibly displaced, and later became seasonal workers."

A sociologist with a degree and a master's in the field, who carried out extensive fieldwork from 2009 to 2015 in Adana, Mersin, Manisa, Bursa and Cizre and is now working on a doctorate, studies rural transformation and migrant labour. Methodological reflexivity: political-economic geography determines where ecological problems will emerge and become visible. Since the 1990s, rural transformation has unfolded at three levels: seasonal agricultural worker profiles have changed — from small farmers supplementing income to entirely landless, entirely vulnerable workers; producers have been transformed — under the new conditions they can no longer produce as they once did; the spatial geography has shifted — labour has been moved to different regions, different conditions.

In agricultural labour there is an ethnic hierarchy: Turkish workers receive the highest wages and Kurdish workers middling ones — some of them left landless by the forced displacement of the 1990s — while Syrian refugees work for the lowest wages, sometimes for none at all. This is not coincidental but structural — the product of Turkish state policies (assimilation, forced displacement) and global refugee crises.

Geo-spatial mapping of labour patterns and migration flows is carried out with QGIS. The concept of "rural ghetto" emerges — permanent tent settlements in Adana-Mersin for Syrian and displaced Kurdish populations. These are not temporary seasonal camps; they create trapped communities with no way out, people who live there all year round. The Seasonal Worker Migration Communication Network has met twice a year since 2010 — researchers, activists and civil society workers from different disciplines. The Rural Research Network has just been founded; a migration network attached to a research institute also holds seminars.

Child labour, gender wage gaps, dispossession — the vulnerability created by landlessness — these are the realities that ecological agriculture research ignores. Ecology cannot be understood without labour, labour without ecology — these are not separate struggles.

The question remains open: how can we understand ecology without understanding labour?

THE TRUST RELATION

A music teacher, a voice that turned down employment in both the state and the private sector, whose environmental consciousness was shaped by the movement against hydroelectric plants and by the Alakır Valley experience — a journey that evolved from a permaculture course into food community work.

Kadıköy Cooperative: builds a direct relationship between producer and consumer. Not only environmental sustainability but also relational justice is kept in view — working conditions, refugee employment, gender dynamics in agricultural households are regularly monitored and evaluated. Ekorita: an interactive ecological map combining ecological spaces, forums, news and recommendations — an answer to information fragmentation. Zero Waste Platform: going beyond recycling to refuse waste at its source, in production — a transformation of consumption, not merely waste management.

> "There's a lot of intention, but when it comes to actually being active, we're a bit sluggish right now."

Food communities and cooperatives operate through horizontal organisation — consensus-based decision-making, no hierarchy, equal participation, transparency. Monthly meetings in person create solidarity, reduce carbon footprint, establish trust relations. Small-scale, distributed networks are more resilient and politically more meaningful than centralised NGOs. The trust relation — not a contract, but a face-to-face relationship — is the foundation of the

alternative economy.

But volunteer burnout is real. Scaling up without compromising principles is difficult. Because it runs on volunteers, the Ekorita project cannot currently move at the pace hoped for. Intentions are many, sustainable action is scarce — this is not only an individual but a structural problem. Volunteer-based sustainability is a structural weakness: people burn out, projects stop, restarting becomes a little harder each time. Without material conditions — time, space, income security — the ideal of horizontal organisation never touches the ground.

The journey from music teaching to permaculture, from resisting hydroelectric plants to the food cooperative, is the story of how individual transformation can evolve into social organisation. Meeting somewhere once a month — so simple an act, carrying so deep a meaning.

THE GREATEST MICROBES

For a voice born in Ordu and raised amid the ecological richness of the Black Sea, settling in Ankara to study at university was an experience of displacement. This experience is the starting point of a soil art practice. Master's research on soil art in Turkey in 2014, scientific illustration training, a doctorate on body-nature dialogue.

Pelisiyar Contemporary Art Initiative (2013): interventions in abandoned and ecologically transformed historic spaces — the space itself becomes material and message. "The Greatest Microbes" — an ongoing performance/street art project. Inspired by a metaphor from a Chinese author: if microbes had hands and spray paint, they would write "The Greatest Microbes" everywhere. At 350 sites in Paris, spreading to Ordu, Ankara and Konya. Microbes are the ultimate survivors — they continue to live even after nuclear devastation. We are transient, they are permanent.

> "Even if humanity ceases to exist on this planet we inhabit, they will continue to live."

An evolution from seeing nature and art as separate, to understanding that art history has included land art (since the 1960s), to scientific illustration deepening knowledge of the flora of Turkey, to performance becoming bodily ecology. The body is not an observer standing outside the landscape but a biological part of it. Most artists concerned with ecology do not deeply feel natural processes — one has to approach them with a farmer's perception: continuous observation, careful analysis, deep emotional connection.

The D8M project: a collaboration with a bulldozer firm in Istanbul. Research on garden and landscape culture in Paris, academic teaching (in Konya, away from Ankara). Moving into action — a transition from hope-based activism to movement-based activism. One does not hope, one moves. Movements — not singular but plural, not linear but dynamic — continue. People speak of space colonisation, of farming on Mars — but even if humanity were to disappear from this planet, microbes would continue to live. Human exceptionalism may be the greatest obstacle to ecological thinking.

PRODUCERS' MARKETS

A designer trained in architecture, with a master's in digital tectonics, who teaches at a university and researches digital production methods — a design researcher working on craft, tools and learning through making. The Menderes Basin agricultural markets (Ödemiş, Tire, Nazilli, Karacasu) were examined as part of research carried out for a design biennial — together with an architect who researches markets and an agricultural engineer who is also a gastronome. Production-consumption networks constitute urban-rural relations — understanding these networks means understanding the food system.

Mapped in the 2017–2018 fieldwork: seasonal change and social function of producers' markets, supply chains, market stall aesthetics (coloured tarpaulins and covers signalling product type), producer profiles, intergenerational knowledge, infrastructure, profit margins.

Vildan Teyze in Tire: a 40-hour labour cycle — garden preparation, market setup, sales — for minimal profit margins. These 40 hours expose the hidden labour behind the discourse of "local food." How deep is our consciousness as consumers of local and organic food? The distance between the hand that buys the product and the hand that grew it is not only physical but epistemological. Neither a production-side nor a consumption-side analysis is sufficient — relational analysis is central. Markets are third spaces where producer skill (curatorship of abundance) meets the consumer's willingness to engage.

Different scales reveal different relations: producer logic at field level is fundamentally different from market presentation. Visual documentation on Instagram was deliberately used — to shift the conversation from "what is this?" to "where does it come from? do you know who grew it? under what conditions?" Coloured tarpaulins signal product type, seasonal variety reflects social function — the market is not merely a place to shop but a place to exchange knowledge. The producer-consumer relation creates — or can create — a trust network beyond market logic.

THE ARCHITECTURE OF SOIL

An architect born in Mardin, whose family goes back to a deep-rooted Sufi tradition, who moved from madrasa education to formal schooling, and who grew up with the family's agricultural knowledge and the landscape stewardship of the Sultan Şeyhmus region (25 degrees against Mardin's 40; a rare green zone). The burning of the villages and forced displacement in 1993, during the Kurdish conflict — the family moved from the village to the district town of Kızıltepe. A witness to childhood friends becoming seasonal agricultural workers.

A TÜBİTAK wind turbine acceleration project (secondary school), two patents, student activism, mathematics and chess competitions — a multifaceted mind. A housing proposal for seasonal agricultural workers in Sarıcakaya, Eskişehir — a micro-climate zone where asparagus is grown. Analysing the spatial evolution of worker settlements between 2002 and 2017 with Google Earth — fifteen years of change, readable in satellite images. A mud-brick architecture proposal inspired by the Harran houses and by the passive cooling technique of the malkaf (wind tower) — multi-layered systems using soil as the primary insulation material. The master plan: shared cooking area, social spaces, permaculture integration — workers growing their own food (tomatoes, aubergines, peppers). Dignified housing, healthy living, organisational capacity — these are directly connected to architectural decisions.

> "Rural architecture is a sustainable ecology too. Mardin is architecture without architects, a city that settled itself. We know that the best insulation is earth."

Architecture is not separate from ecology. Housing design directly affects the dignity, health and organisational capacity of agricultural workers. Seasonal workers live in tents, shacks, containers — these spaces draw not only physical but social boundaries. A dignified living space is a prerequisite for organising.

The transition from hazelnut to kiwi — a transformation lived through in one family — brought new harvest seasons, new recipes, new social practices, and changed the rhythms of the community's life. Has a kiwi culture formed? How long does culture take to form? When agricultural policy eliminates a crop, it also erases the culture and knowledge systems bound to that crop — the family economics of hazelnut picking, community rituals, seasonal order disappear. Crops are not economic units; they are cultural vectors, carriers of knowledge, embodied forms of social relations.

The family's agricultural knowledge of the Sultan Şeyhmus region — knowledge transmitted across generations, learned through living — was severed with the forced displacement of 1993. This severance is not only geographic but epistemological: when knowledge is displaced, like a seed torn from its soil, it dries up.

ROMA BOSTANI AND SEEDS

A materials engineer who worked five years in media, who experienced an "awakening" after Gezi — particularly after learning permaculture — and chose to leave corporate work to move towards urban food production and self-sufficiency.

Roma Bostanı — a community garden — is a reclamation of public space against the municipality's attempt to convert it into a commercial café, proof that urban agriculture is possible. Soap-making — transforming personal consumption into natural home production. Seed guardianship — multiplying and distributing local seeds, a newly begun practice. Food community organising.

The permaculture framework: how can we meet our needs with minimum harm to nature and how can we convert outputs into inputs? For an urban dweller, disconnection from production is not destiny — consumption dependency can be reduced through knowledge of materials, direct production and sharing networks. Making soap, saving seeds, setting up a garden — these seem like small acts but each is a point of disconnection from the system. At Roma Bostanı the legal battle was won, but the complacency that followed is a reminder that engagement has to be sustained. Winning is not enough; what is won must be protected.

> "Food is very important. Our local production has dropped incredibly. What can we do locally?"

This is the question that guides the whole session.

THE BREATH OF SOIL

This is the ninth and final session of birbuçuk's Solunum programme. Over more than a year and a half, a journey has been completed: from water to biodiversity, from metabolism to boundaries, from climate to mining, from gender to energy, and on to soil. Every session brought people from different disciplines to the same table — no hierarchy, equal time, personal narrative, formal structurelessness.

The soil session is both the summary of this journey and its test. Seven presentations — food community organiser, agricultural activist, labour sociologist, food cooperative volunteer, soil artist, architect-researcher, soil architect — have touched the same matter from different paths. And in free discussion these voices have mingled, complemented and sometimes contradicted one another. But the fundamental consensus that emerges is clear: soil is a living system, labour cannot be separated from ecology, trust relations are the foundation of the alternative economy, knowledge systems are erased together with crops and soil practices.

The depoliticisation of ecology — "For Nature" concerts, sustainability branding — conceals systemic causes. Discussing climate change while smoking a cigarette is to avoid making the personal-political connection. Design and sustainability are widely discussed in academia but rarely translated into practice. Conversely, activist and practitioner knowledge rarely reaches academic contexts. Three-hour sessions are insufficient for sustained organising — follow-up workshops, small working groups, documentation are needed.

Dominant tensions: between urgency and patience — the slowness of relationship-building while the climate crisis accelerates. Between systemic critique and gradual change — how to stay motivated? Between the fragmentation of knowledge across disciplines and the need for a common framework. Scale: individual actions are insufficient, structural change is necessary but seems impossible.

Most projects work at intermediate scales: not global policy, not individual consumption, but neighbourhood and regional networks — markets, communities, collaborative workshops. Change arises from accumulated small practices and local organising — not from top-down implementation. The agricultural activist's fifteen years of commitment to organising, the food community organiser's nearly four years of work on Roma Bostanı, the labour sociologist's multi-year ethnographic immersion — these time scales require emotional resilience and material security that most people lack.

But soil teaches us something: when the first drop of rain falls on the dry surface, microorganisms awaken. The conditions do not need to be perfect for awakening — one drop is enough. Movement — not singular but plural, not linear but rhizomatic, not centred but distributed — takes place simultaneously through many practices, at different scales, in different geographies. The Solunum programme has itself been this practice: it has created unexpected continuities and relations, people from different disciplines have learned to ask the same questions in different languages, participants have realised they will continue working in overlapping fields. From the community garden to the producers' market, from the seasonal worker tent to soil architecture, from the world of microbes to the food cooperative — all are nodes of the same network.

Documentation, publication and future workshop series have been proposed — production-focused workshops, panels, film screenings, small exhibitions. Written outputs — the text of each session, and a book that will also include the research. This session is not an ending but a node in an ongoing network.

Socio-economic metabolism — how we as communities organise our surroundings, input from outside, processing inside, output to outside. Unity of knowledge — not remaining bound to a single discipline, looking at the whole. Rhizome — networks without a centre, multiplying horizontally, continuing even when broken. These three concepts are the core of birbuçuk's Solunum programme, and in the soil session they were tested for the last time, but in their most concrete form. Like soil: cyclical, living, awakening with the first drop even when it looks dry.